What is Software Architecture and Design?

SYSC 4120: Software Architecture and Design

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Winter 2025

Material Based On: *Fundamentals of Software Architecture* (Chapter 1 & 2) by M. Richards and N. Ford

Lecture Objectives

1. Define software architecture and software design
2. Describe the importance of software architecture and design in the software engineering lifecycle

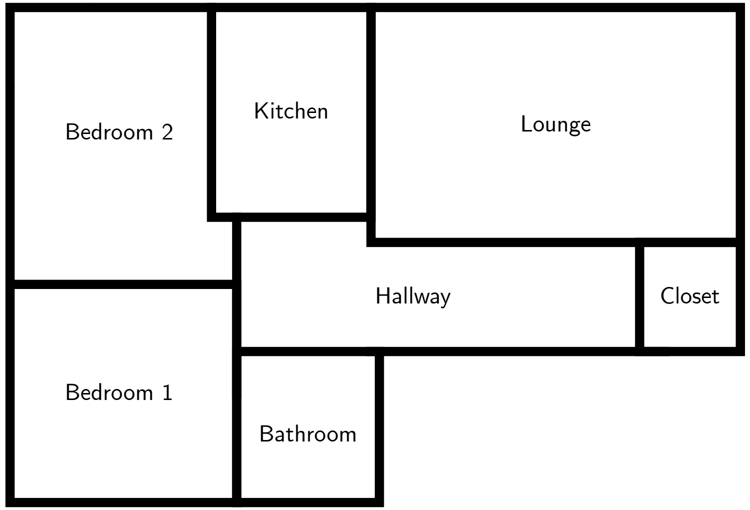
What is Architecture and Design?

A Design Example

Example (Moving into a New House)

Consider the actions that might be involved when planning a move into a new house.

1. Measure the dimensions of the various rooms and determine locations of doors, windows, etc. and draw a scale floor plan
2. Measure the dimensions of various items of furniture and to cut out cardboard shapes to represent these on the chosen scale

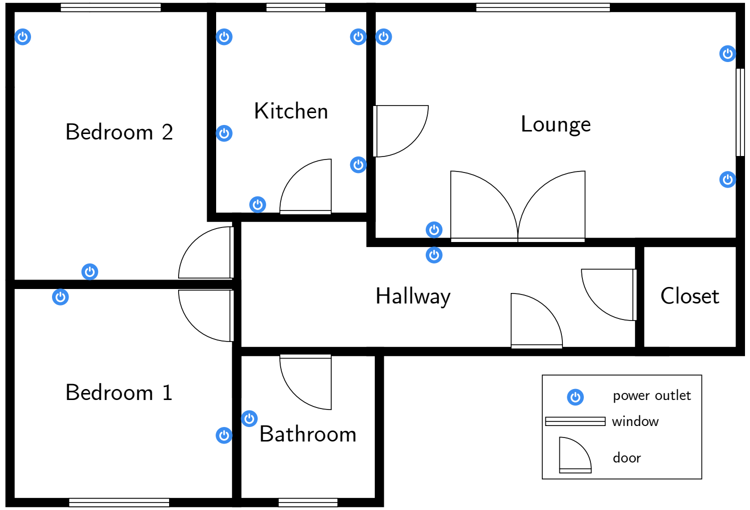


A Design Example

Example (Moving into a New House (continued))

The *design process* then involves trying to identify the 'best' positions for the pieces of card on the plan of the new house. There is really no analytical form that can be used to do this, since it is usually necessary to trade off different factors in making the choices.

- **Strategies**: e.g., determine which rooms will take greatest priority
- **Constraints**: e.g., power outlet requirements, functionality, style



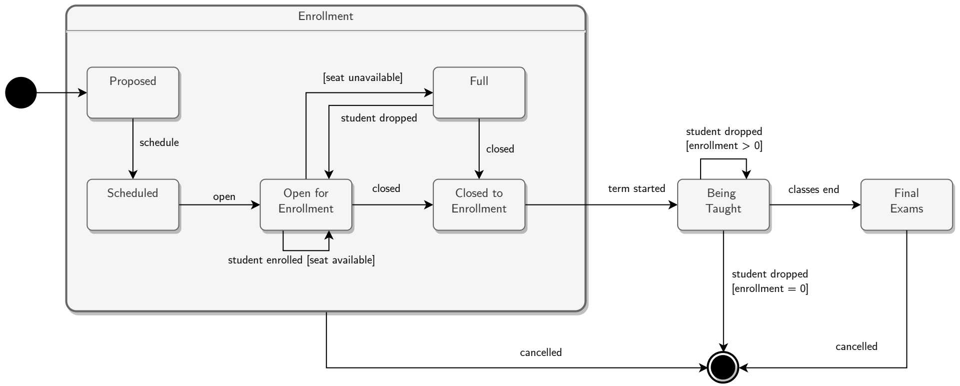
What is Design?

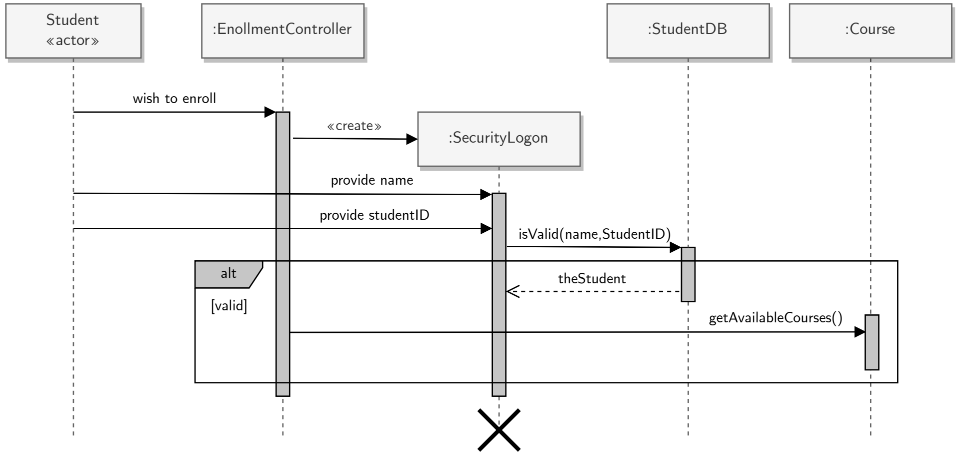
- The nature of design is not easy to grasp in a clear definition
- The single word "design" encompasses an awful lot of objective (functions) and subjective (aesthetic) aspects
  - Usually requires considerable research, thought, modeling, interactive adjustment, and re-design
- Depending on the designed entity, either the objective or the subjective aspects might takeover
  - *When the subjective aspects take over:* design is viewed as a more rigorous form of art (art with a clearly defined purpose)
  - *When the objective aspects take over:* design is viewed as a transformation expressing the functional requirements into a collection of entities with their connectors

What is Software Design?

Software Design

Software design is the process by which an agent creates a specification of a *software artifact* intended to accomplish goals, using a set of primitive components and subject to *constraints*.





Revisiting the Design Example

Example (Building a New House)

Suppose instead of moving into a house, we want to build our dream home. When designing a house that will be liveable we must consider several constraints that influence our design decisions:

1. Look and Feel
2. Functionality
3. Structural integrity
4. Budget and Timeline
5. etc.

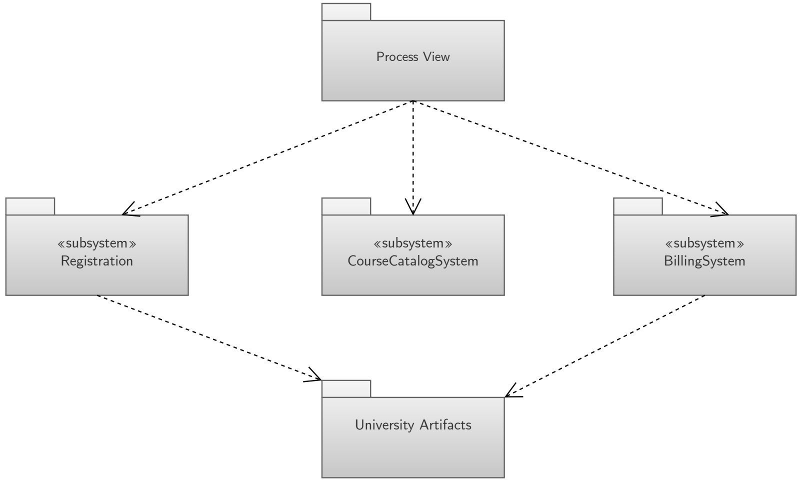


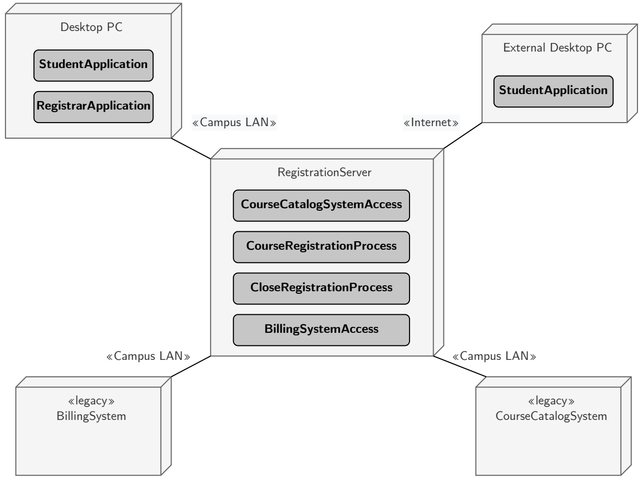

What is Software Architecture?

Software Architecture

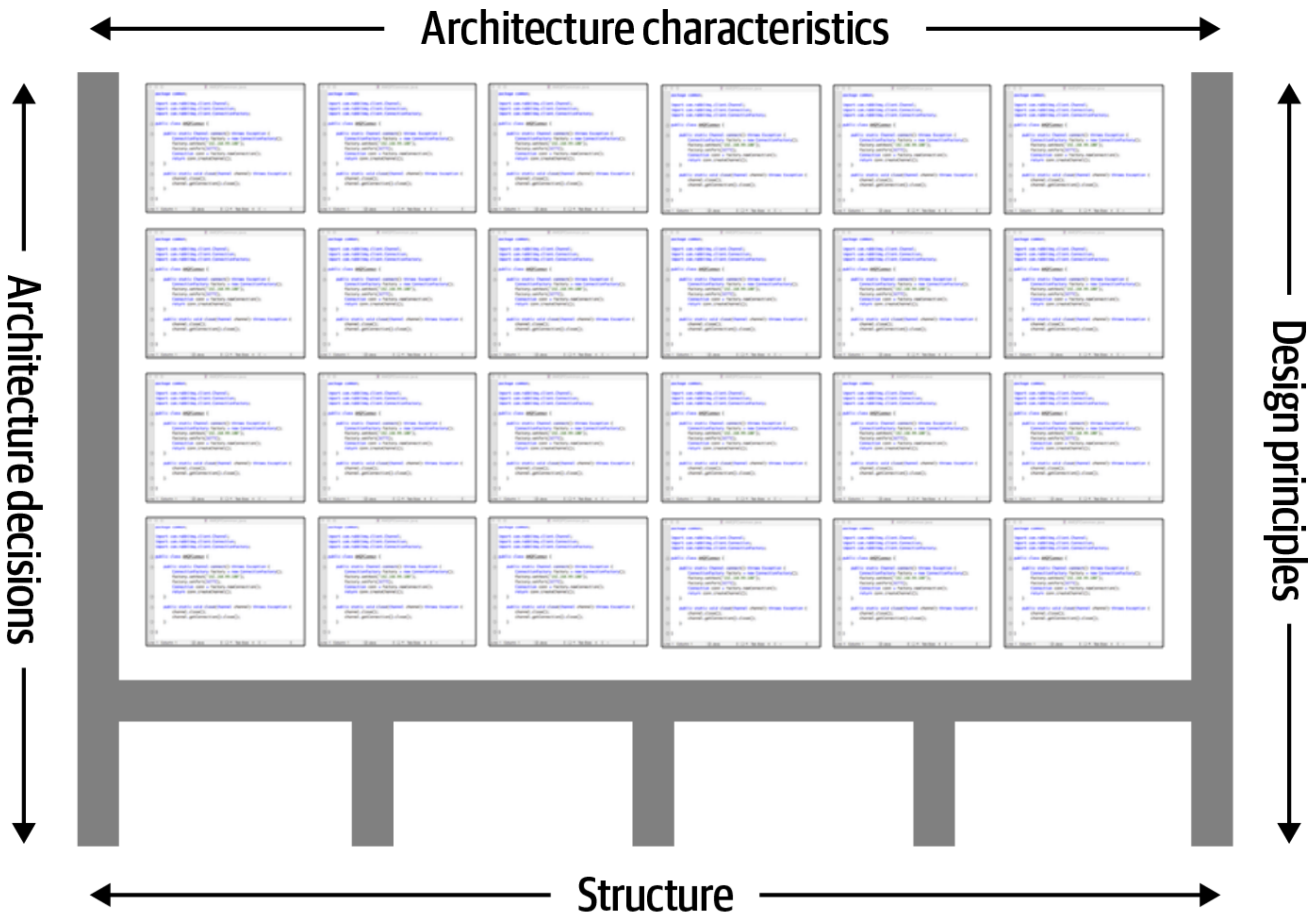
The set of structures needed to reason about the system, which comprise software elements, relations among them, and properties of both.

- Architecture is a set of software structures
- Architecture is an abstraction
- Architecture is a moving target





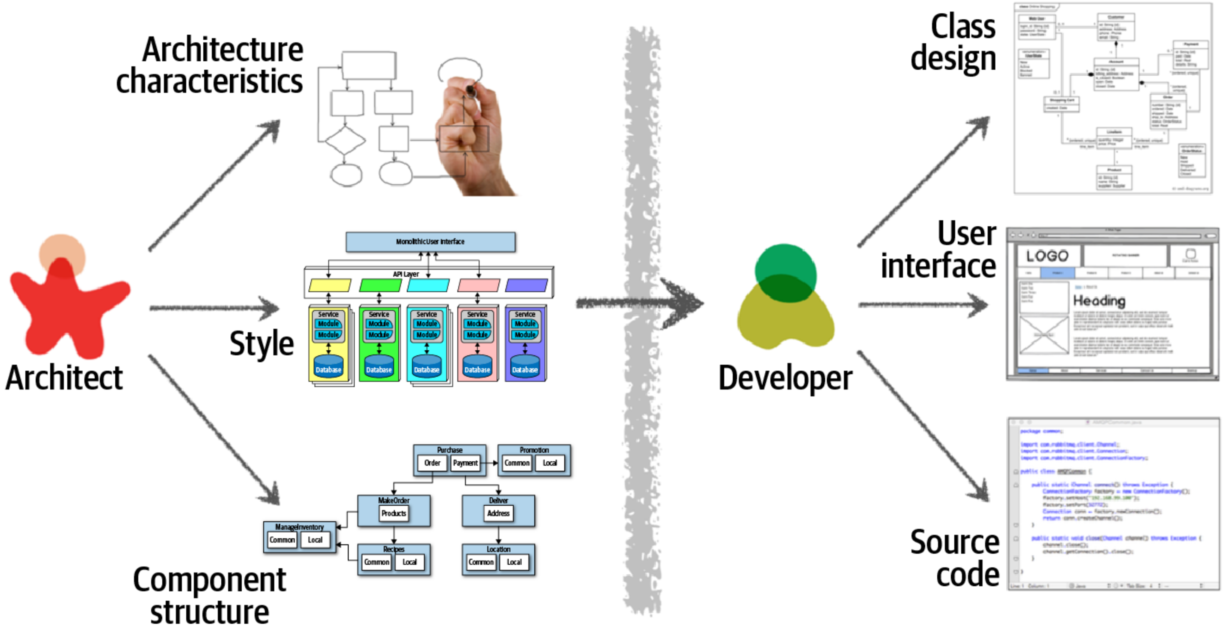
Thinking About Software Architecture



Architecture vs. Design

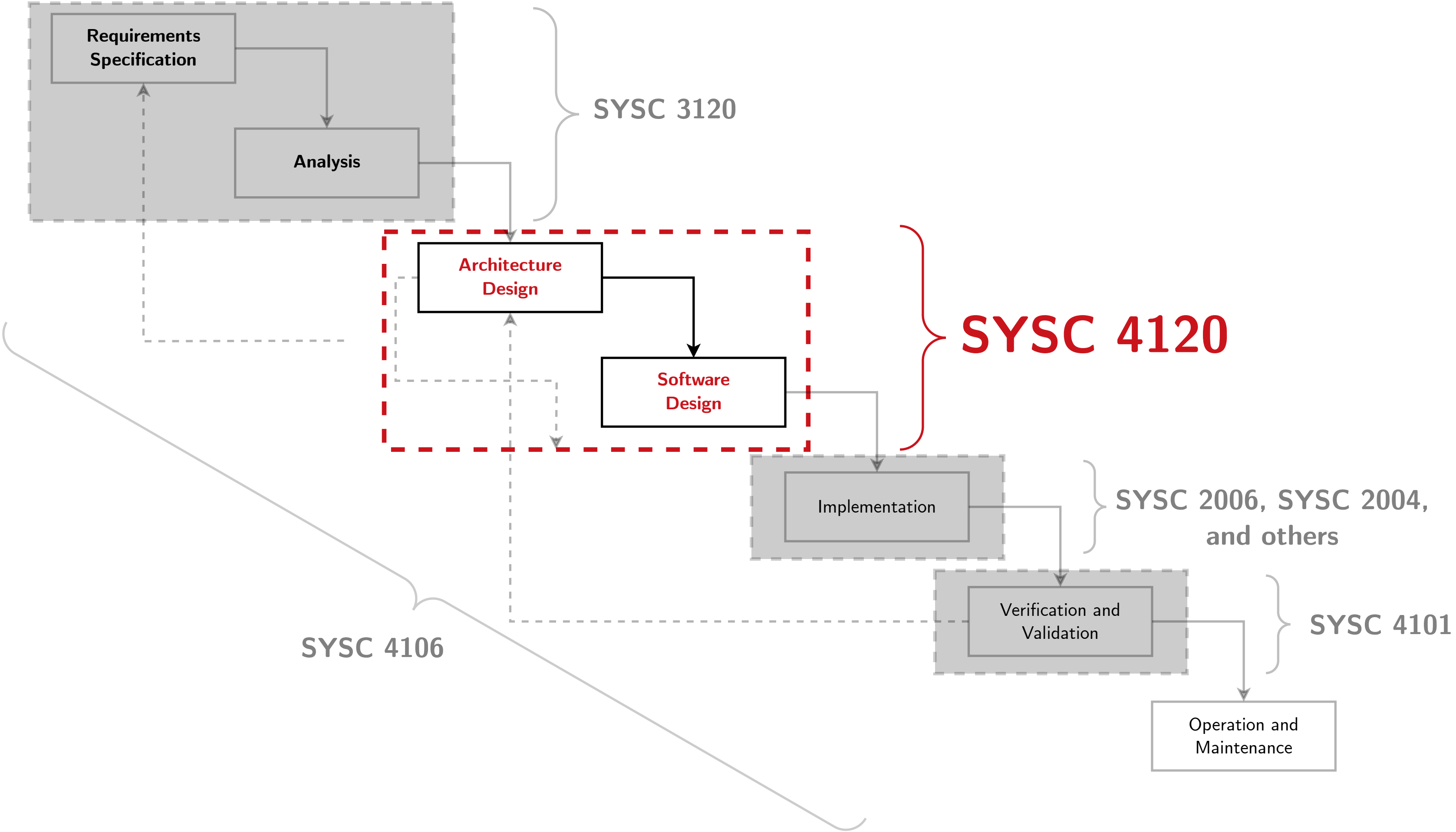
Architecture is design, but not all design is architecture!

Many design decisions are left unbound by the architecture and depend on the discretion and good judgment of downstream designers and even implementers.



Why Software Architecture and Design?

Focus of SYSC 4120



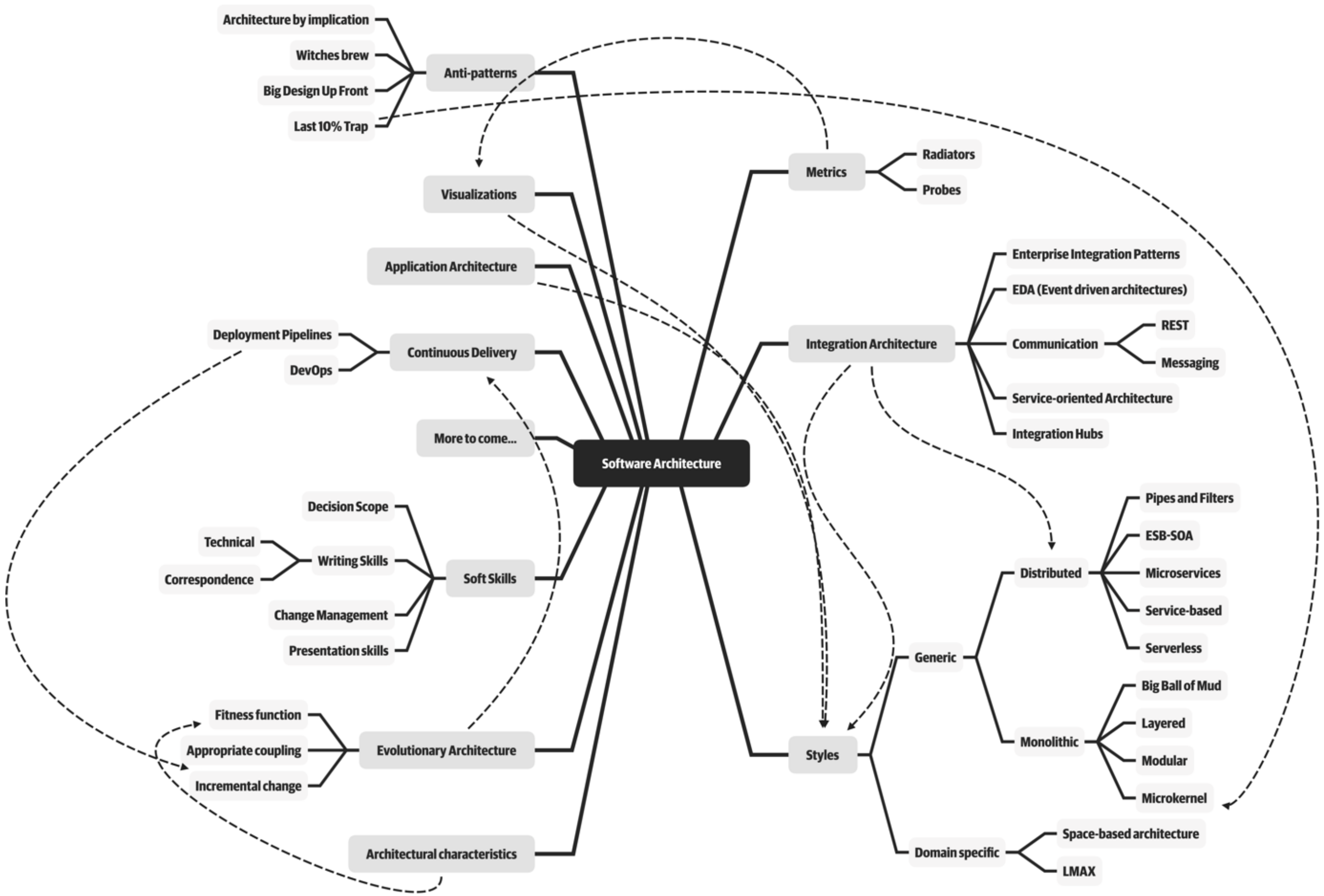
Why is Software Architecture Important?

1. An architecture can either inhibit or enable a system's driving quality attributes.
2. The decisions made in an architecture allow you to reason about and manage change as the system evolves.
3. The analysis of an architecture enables early prediction of a system's qualities.
4. A documented architecture enhances communication among stakeholders.
5. The architecture is a carrier of the earliest, and hence most-fundamental, hardest-to-change design decisions.
6. An architecture defines a set of constraints on subsequent implementation.
7. The architecture dictates the structure of an organization, or vice versa.

Why is Software Architecture Important?

8. An architecture can provide the basis for incremental development.
9. An architecture is the key artifact that allows the architect and the project manager to reason about cost and schedule.
10. An architecture can be created as a transferable, reusable model that forms the heart of a product line.
11. Architecture-based development focuses attention on the assembly of components, rather than simply on their creation.
12. By restricting design alternatives, architecture channels the creativity of developers, reducing design and system complexity.
13. An architecture can be the foundation for training of a new team member.

Responsibilities of Software Architects and Designers



Laws of Software Architecture

First Law of Software Architecture

Everything in software architecture is a trade-off.

If an architect thinks they have discovered something that isn't a trade-off, more likely they just haven't identified the trade-off yet.

Second Law of Software Architecture

Why is more important than how.

Lecture Summary and Problems

Lecture Summary

- Provided definitions of software design and software architecture
- Discussed the importance of software architecture and design in the software engineering lifecycle

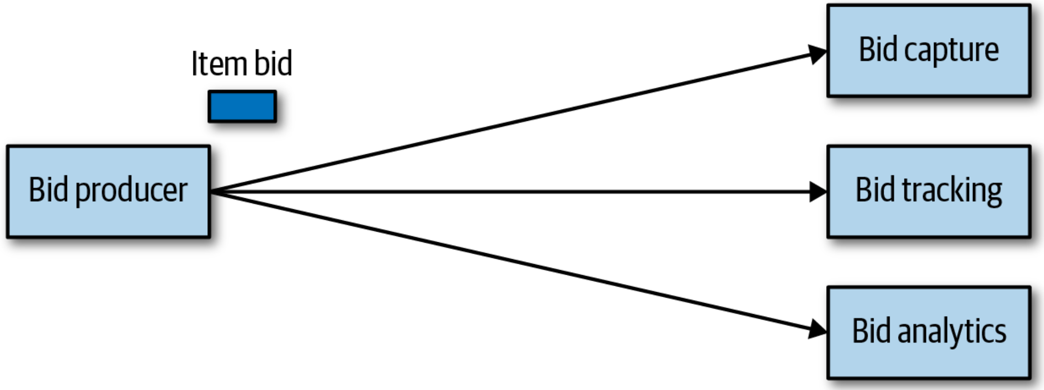
Check Your Understanding!

Exercise (Analyzing Trade-offs)

Consider an item auction system, as illustrated in below, where someone places a bid for an item up for auction. The `Bid Producer` service generates a bid from the bidder and then sends that bid amount to the `Bid Capture`, `Bid Tracking`, and `Bid Analytics` services. This could be done by:

- using a topic in a publish-and-subscribe messaging fashion
- using queues in a point-to-point messaging fashion

Which one should the architect use?



References & Further Reading

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